



LINESIGHT QC · TECHNICAL SYNTHESIS

Edge AI quality pipeline for MSME production lines

Hardware and software synthesis of DFRobot HuskyLens 2 (6 TOPS) on D-Robotics RDK X5 8GB (10 TOPS BPU), with LineSight as the operator console.

SENSOR

HuskyLens 2
Kendryte K230 · SEN0638

ORCHESTRATOR

RDK X5 8GB
Sunrise 5 · Ubuntu 22.04

CONSOLE

LineSight
Pass / fail / hold · CAPA

1. Purpose

MSME shops (textiles, bottled FMCG, stamped metal, SMT, ceramics, carton print) need 100% visual gates without a dedicated AOI cell. LineSight splits work three ways:

- **HuskyLens 2** sees the unit — onboard YOLO, classification, instance segmentation, OCR, barcode, and self-learning.
- **RDK X5 8GB** owns policy — pixel-to-mm, spec limits, AQL sampling, GPIO / CAN reject.
- **LineSight** is the operator surface — lot genealogy, override, teach set, on-demand diagnosis.

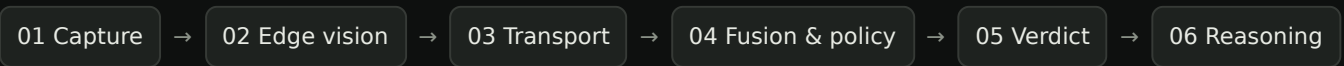
This document is the synthesis of that pipeline: devices, stages, protocols, MSME recipes, and the software that the live console implements. The in-browser LineSight run is a faithful demo of the chain (simulated HuskyLens feed). Plant deployment uses the same stages over USB 3.0 / UART against real hardware.

2. Hardware stack

SENSOR		ORCHESTRATOR	
HuskyLens 2		RDK X5 8GB	
SKU	DFRobot SEN0638 / SEN0638-TF	Board	D-Robotics RDK X5 8G
SoC	Kendryte K230 dual-core 1.6 GHz	CPU	8× Cortex-A55 @ 1.5 GHz
NPU	6 TOPS	BPU	Bayes 10 TOPS (INT8)
Memory	1 GB LPDDR4 · 8 GB eMMC/TF	GPU	32 GFLOPS
Imager	GC2093 2 MP, 1/2.9", 60 fps	RAM	8 GB LPDDR4
Panel	2.4" IPS 640×480 capacitive	OS	Ubuntu 22.04
I/O	USB-C, Gravity 4-pin UART/I2C, TF	Cameras	2× 4-lane MIPI CSI (optional 2nd view)
Power	3.3-5 V · 1.5-3 W	Host	4× USB 3.0 Type-A (HuskyLens)
Software	20+ models, custom YOLO, MCP server	Actuate	GPIO, CAN FD reject / press interlock
		Net	GbE, Wi-Fi 6, BT 5.4

Combined edge compute: 6 TOPS (K230) + 10 TOPS (X5 BPU) = 16 TOPS on the line, no cloud round-trip per unit. Optional second MIPI camera on the X5 covers top + side without a second HuskyLens.

3. Six-stage pipeline



01	Capture GC2093 2 MP at 60 fps. Fill LEDs on the HuskyLens 2; optional microscope lens for SMT pads and print register. Exposure held constant per recipe so shade ΔE and fill-level pixels stay comparable across a lot.	GC2093 2 MP
02	Edge vision K230 runs the recipe's model mix locally: object recognition, classification, instance segmentation, color, OCR, barcode/QR, object tracking, and the self-learning classifier. Output is blocks (xywh, label, confidence) plus optional MCP semantic frames — not a raw video dump.	HuskyLens 2 · 6 TOPS
03	Transport USB 3.0 for stills / MJPEG when needed; Gravity UART (115200, HuskyLens V2 protocol 0x55 0xAA) for blocks and arrows. Typical inspect-path latency 5–8 ms on the demo line. Zero drops is the gate for a production link.	USB 3.0 / UART 115200
04	Fusion & policy RDK X5 applies homography (pixel → mm), spec min/max, hole-count / fill-ml calibration, AQL skip-lot, and SPC. BPU can fuse a second MIPI view. Reject pulse on GPIO or CAN FD to the solenoid / press. This is where MSME mix lives: one image, many SKUs.	RDK X5 BPU · 10 TOPS
05	Verdict LineSight stamps pass fail hold . Operator may override. Unit joins lot genealogy with cycle time, score, detections and measurements. Hold is for borderline confidence, not a silent pass.	LineSight console
06	Reasoning On operator request only: Grok reads the fail packet (recipe, detections, specs) and returns Cause / Process / Operator / CAPA. If the xAI key is quota-blocked, an on-device policy note runs the same four-line shape from the recipe rules. Never called per frame.	Grok MCP · on-demand

4. Protocol and data

HuskyLens 2 binary frame (V2 protocol):

0x55 0xAA | CMD | ALGO_ID | LEN | DATA | CHK

Algorithm IDs used on this line: object recognition (2), tracking (3), color (4), classification (5), self-learning (6), instance seg (7), OCR (11), barcode / QR / tag. LineSight consumes the decoded blocks as:

```
{ id, label, x, y, w, h, confidence, kind, severity? }
```

Measurements are policy, not camera: each recipe declares specMin / specMax / target. A unit fails if a critical defect block fires or any measurement is out of spec. Score starts at 100 and is penalised per defect and OOS channel.

5. MSME inspection recipes

SECTOR	SKU	CYCLE	HUSKYLENS ALGOS	RDk X5 ROLE	CRITICAL DEFECTS
Textiles Woven fabric	TX-W0V-240	420ms	Instance seg, color, classify	Shade ΔE vs master, defect area vs GSM	Pin hole, weft stain, shade band
Food & beverage Bottled FMCG	PK-BTL-500	280ms	Object, OCR, color, barcode	Fill px→ml, cap GPIO reject	Underfill, missing cap
Metal fabrication Stamped bracket	MT-BRK-08	360ms	Object, instance seg, classify	Homography, hole count, CAN to press	Missing hole, length OOS
Electronics SMT board	EL-PCB-12A	510ms	Classify, object, self-learn	BOM overlay, polarity map, hold for rework	Missing 0805, IC polarity
Ceramics Glazed tile	CR-TIL-300	390ms	Instance seg, color, classify	Crack length vs class, kiln feedback	Hairline crack, edge chip
Packaging print Carton print	PR-CTN-A4	310ms	OCR, barcode, QR, object	GS1 grade, colour-to-colour register	Unreadable barcode, wrong batch text

Sampling examples: 100% fill + cap with 1/10 OCR audit (FMCG); 100% holes with 5-pc/hr CMM audit (metal); 100% first-article then 100% AOI (SMT).

6. LineSight software map

Surfaces

- **Station** — live HuskyLens viewport, pipeline rail, verdict, diagnose
- **Pipeline** — hardware topology and this synthesis
- **Lots** — yield, score trend, defect Pareto, genealogy
- **Recipes** — six MSME SKUs, load onto the station
- **Teach** — gold / defect samples → K230 self-learn deploy

Runtime rules

- No accounts. Session state in memory (demo lot seeded).
- Grok is user-initiated, 500-token cap, never per frame.
- On 403 / missing key: on-device Cause / Process / Operator / CAPA.
- Self-learn deploy gated at 4 gold + 4 defect (8+8 preferred).
- Demo line is labelled as such — not a fake plant handshake.

7. Plant wiring (when hardware is on the bench)

1. Power HuskyLens 2 from 5 V. Load the recipe model set (or custom YOLO from Model Hub).
2. USB 3.0 Host on RDK X5 ← HuskyLens 2 Type-C. UART 115200 on Gravity as fallback / block stream.
3. RDK X5 Ubuntu 22.04 service: decode protocol, apply recipe JSON, pulse GPIO/CAN on fail.
4. LineSight on the same board (or a shop PC) over HTTP to that service.
5. Reject solenoid / tower light on GPIO; press interlock on CAN FD for metal.
6. Teach gold/defect from the first-article lot; deploy to the K230 self-learning head.

Estimated station BOM (indicative, 2026): HuskyLens 2 ~USD 90, RDK X5 8GB board, 24 V reject solenoid, enclosure, fill lighting. Well below a packaged AOI cell, which is the MSME constraint.

8. Why the split holds for MSME

On-device first

Classification and OCR stay on the K230. Bandwidth and privacy stay on the line. Cloud is not in the cycle-time path.

Board does policy

Recipes, AQL and the reject pulse live on the X5. Swap SKU without retraining the whole stack. High mix, short runs.

LLM on demand

Grok reads a fail packet only when an operator asks. Quota stays bounded. CAPA language matches a setter, not a data scientist.

9. Open items for a live cell

- Calibrate homography per nest (checkerboard still on first article).
- Map GPIO bit to the actual reject valve / tower; debounce 3-fail-in-10 as a line stop.
- Balance teach sets (minimum 8 gold + 8 defect per class) before raising confidence thresholds.
- Persist lots to the plant MES or a local Postgres when accounts exist; this demo does not.
- Wire HuskyLens MCP to Grok only on the shop VLAN, never the public internet per frame.

LineSight LS-SYN-001 · Pipeline synthesis · HuskyLens 2 × RDK X5 8GB × MSME QC

Prepared for shop-floor review. Demo console implements stages 01–06 against a simulated feed.